

Robert Hart

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AI-native full-stack engineer — shipped a 20-player real-time multiplayer game, an AI prototype for a startup client, and a multi-agent tutoring platform, all in 2026. 15 years of client-facing work in coaching, research, and creative direction.

Software Engineering Fellow — Fractal Tech, NYC

Feb – May 2026 · Full-time fellowship (700+ hours; six-day weeks)

- Shipped full-stack and real-time multiplayer web apps in rapid sprint cycles (React, TypeScript, Express, MongoDB; WebSocket broadcast patterns with server-authoritative state) — deployed to Render, one containerized with Docker
- Led systems integration on a 4-person team: diagnosed central coordination failure, then designed shared workflows that more than doubled team output on a week-long game build
- Built and tuned multi-agent AI workflows: parallel dispatch, phase-sequenced builds, handoff protocols, context engineering

AI Engineering Intern — Stage 11 Agentics, NYC

Apr 2026 · Two-week engagement, working directly with the founder

- Shaped an open-ended brief into a focused deliverable; shipped a working prototype in two weeks, under cost constraints
- Owned the full delivery cycle: implementation, code-review response, production deploys, QA, and onboarding docs

Webmaster (retainer) — MacAvoy Foundation · May 2026 – Present · Part-time

- Sole technical steward for the nonprofit's web presence: on call for issues and changes, advising non-technical leadership

Immersive Theater Designer & Director · 2024 – Present · NYC

- Directed teams of 4–26 performers in live branching-narrative productions for up to 75 simultaneous participants — interlocking storylines, live improv, audience-driven outcomes; new productions in development
- Co-designed the real-time state management systems tracking player and faction status across 5-hour productions

Freelance Vocal Coach & Creativity Coach · 2023 – Present · NYC

- Run own client practice (30+ clients): find real needs, then design individualized developmental programs — specializing in stage-fright intervention and embodied self-awareness
- Built pedagogical frameworks grounded in developmental psychology (Kegan, Piaget) that now drive Weft's scaffolding logic

Lead Researcher — DesignIntelligence · 2020 – 2021 · Atlanta/Remote

- Owned the research-to-publication pipeline for strategic reports advising C-suite leaders in architecture, engineering, and construction: primary research (interviews and surveys with dozens of industry professionals), quantitative analysis (healthcare economics, demographics, market forecasting), and all writing — delivered on fixed 60–90-day cycles
- Designed the firm's research methodology and each report's evaluative frameworks, reconciling conflicting sources into decision-grade guidance for non-technical executives
- Built out a nascent research department: recruited and vetted researcher candidates, hired one, and trained and mentored two — editing their reports and developing their projects from initial question to published deliverable

Earlier — rapid-study track record · Insurance agent, 2009 · Placement-exam tutor, 2017 – 2021

- Self-studied 100 hours in 10 days to pass insurance licensing exams, then translated legal-financial products into plain language for customers daily; as a tutor, mastered multiple subjects and taught across skill levels, often in one classroom

Projects

Weft — Developmental AI Tutoring Platform | 2026 – present · TypeScript · Claude Agent SDK · [code](#)

- Designed and built a developmental AI tutoring system with 10+ agent skills encoding pedagogical logic: gap-type classification, learner state tracking, intervention matching, and spaced review
- Architected multi-agent orchestration — parallel dispatch, phase sequencing, handoff protocols

The Schelling Point — Real-Time Multiplayer Game | 2026 · React · TypeScript · Express · WebSocket · [play](#) · [code](#)

- Led a 4-person team from an open brief (“a party game of some kind”) to a scalable concept — 20+ players converge on shared words via embedding-space centroids — shipped in a 5-day sprint with a solo overnight push to make demo day
- Redesigned scoring overnight when players competed instead of cooperating; 1st live group of 20 won together

PulseMap — Open Protocol for Synchronized Media Experiences | 2026 – present · TypeScript · SDK design · [code](#)

- Designing an open-source protocol for layering synchronized experiences onto time-based media

Languages & Frontend: TypeScript, JavaScript, React, HTML, CSS

Backend & Data: Node.js, Express, WebSocket, MongoDB, PostgreSQL, SQLite, REST API design

AI & Agents: Claude Code, Claude Agent SDK, multi-agent orchestration, context engineering

Deployment & Tooling: Docker, Render, Railway, GitHub Actions (CI/CD), Git, Vite, Vitest

Education: University of Georgia — BA, English Literature & Music Business